Calculus Cheat Sheet

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Revision 4

Limits

Existence

$\lim_{x\to a} f(x) = L$ exists if $\forall\epsilon > 0 \exists\delta >$
0 s.t. $|x-a| < \delta \Rightarrow |f(x) - L| < \epsilon$.

Properties

$\lim_{x\to a}[cf(x)] = c\lim_{x\to a}[f(x)]$
$\lim_{x\to a}[f(x) \pm g(x)] = \lim_{x\to a}[f(x)] \pm \lim_{x\to a}[g(x)]$
$\lim_{x\to a}[f(x)g(x)] = \lim_{x\to a}[f(x)]\lim_{x\to a}[g(x)]$
$\lim_{x\to a}\left[\frac{f(x)}{g(x)}\right] = \frac{\lim_{x\to a}[f(x)]}{\lim_{x\to a}[g(x)]}, \lim_{x\to a}[g(x)] \neq 0$

Evaluation Techniques

Basics

$\lim_{x\to a}[f(x)] = f(a)$ if f exists at a

L'Hôpital's Rule

If $\lim_{x\to a}\left[\frac{f(x)}{g(x)}\right] = \frac{0}{0}$ **or** $\frac{\pm\infty}{\pm\infty}$ then
$\lim_{x\to a}\left[\frac{f(x)}{g(x)}\right] = \lim_{x\to a}\left[\frac{f'(x)}{g'(x)}\right]$

Derivatives

Definition

$\frac{d}{dx}f(x) = f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$.

Techniques

Sum Rule

$\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$

Product Rule

$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$

Quotient Rule

$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$

Power Rule

$\frac{d}{dx}(x^n) = nx^{n-1}$

Chain Rule

$\frac{d}{dx}(f(g(x))) = f'(g(x))g'(x)$

Parametric

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ recurse y for nth derivative
$(x, y) + (x', y')t$ for parametric tangent

Common Derivatives

$\frac{d}{dx}(x) = 1$
$\frac{d}{dx}(\sin x) = \cos x$
$\frac{d}{dx}(\cos x) = -\sin x$
$\frac{d}{dx}(\tan x) = \sec^2 x$
$\frac{d}{dx}(\sec x) = \sec x \tan x$
$\frac{d}{dx}(\csc x) = -\csc x \cot x$
$\frac{d}{dx}(\cot x) = -\csc^2 x$
$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} \sin^{-1} x \neq \frac{1}{\sin x}$
$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
$\frac{d}{dx}(a^x) = a^x \ln a$
$\frac{d}{dx}(e^x) = e^x$
$\frac{d}{dx}(\ln x) = \frac{1}{x}, x > 0$
$\frac{d}{dx}(\ln|x|) = \frac{1}{x}, x \neq 0$
$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$

Common Chain Rule Derivatives

$\frac{d}{dx}([f(x)]^n) = n[f(x)]^{n-1}f'(x)$
$\frac{d}{dx}(e^{f(x)}) = e^{f(x)}f'(x)$
$\frac{d}{dx}(\ln[f(x)]) = \frac{f'(x)}{f(x)}$
$\frac{d}{dx}(\sin[f(x)]) = f'(x)\cos[f(x)]$
$\frac{d}{dx}(\cos[f(x)]) = -f'(x)\sin[f(x)]$
$\frac{d}{dx}(\tan[f(x)]) = f'(x)\sec^2[f(x)]$
Trig derivatives, same old same old
$\frac{d}{dx}(f(x)^{g(x)}) = \frac{g(x)f'(x)}{f(x)} + \ln[f(x)]g'(x)$

Integrals

Definition

The integral of some function $f(x)$ is a function $f^\star(x)$ s.t. $f^{\star\prime}(x) = f(x)$. **Don't forget your constant!** The integral can be done using the Right Riemann Sum:
$\int_a^b f(x)\,dx =$
$\lim_{n\to\infty}\sum_{i=1}^n\left[\frac{b-a}{n}\cdot f(a + i\cdot\frac{b-a}{n})\right]$

U-Substitution

$\int_a^b f(g(x))\cdot g'\,dx = \int_{g(a)}^{g(b)} f(u)\,dx$. Using U-Substitution, $u = g(x)$ and $du = g'(x)dx$ ($dx = \frac{du}{g'}$).

Integration By Parts

$\int u(x)v'(x)\,dx =$
$u(x)v(x) - \int u'(x)v(x)\,dx$ and
$\int_a^b u(x)v'(x)\,dx = uv|_a^b - \int_a^b u'(x)v(x)\,dx$.

Common Integrals

$\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C$
$\int \frac{1}{x}\,dx = \ln|x| + C$
$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C$
$\int \ln x\,dx = x\ln x - x + C$
$\int e^x\,dx = e^x + C$
$\int a^x\,dx = \frac{a^x}{\ln a} + C$
$\int \sin x\,dx = -\cos x + C$
$\int \cos x\,dx = \sin x + C$
$\int \tan x\,dx = \ln|\sec x| + C$
$\int \sec^2 x\,dx = \tan x + C$
$\int \sec x\tan x\,dx = \sec x + C$
$\int \csc x\cot x\,dx = -\csc x + C$
$\int \csc^2 x\,dx = -\cot x + C$
$\int \sec x\,dx = \ln|\sec x + \tan x| + C$
$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
$\int \frac{1}{\sqrt{a^2+x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right) + C$

Trig Reduction Formulae

(For $n \geq 2$)
$\int \sin^n(x)\,dx =$
$-\frac{1}{n}\sin^{n-1}(x)\cos(x) + \frac{n-1}{n}\int \sin^{n-2}(x)\,dx$
$\int \cos^n(x)\,dx =$
$\frac{1}{n}\cos^{n-1}(x)\sin(x) + \frac{n-1}{n}\int \cos^{n-2}(x)\,dx$
$\int \tan^n(x)\,dx =$
$\frac{1}{n-1}\tan^{n-1}(x) + \frac{n-1}{n}\int \tan^{n-2}(x)\,dx$
$\int \sec^n(x)\,dx = \frac{1}{n-1}\sec^{n-2}(x)\tan(x) +$
$\frac{n-2}{n-1}\int \sec^{n-2}(x)\,dx$

Trig Identities

$\sin^2(x) + \cos^2(x) = 1$
$1 + \tan^2(x) = \sec^2(x)$
$\sin(2x) = 2\sin(x)\cos(x)$
$\cos(2x) = \cos^2(x) - \sin^2(x)$
$= 2\cos^2(x) - 1$
$= 1 - 2\sin^2(x)$

Trig Substitutions

$\sqrt{a^2-x^2} \rightsquigarrow x = a\sin(\theta)$
$\sqrt{a^2+x^2} \rightsquigarrow x = a\tan(\theta)$
$\sqrt{x^2-a^2} \rightsquigarrow x = a\sec(\theta)$

Applications

Centroids

$C = (\hat{x}, \hat{y})$, $\hat{x} = \frac{\int xf(x)\,dx}{\int f(x)\,dx}$ & $\hat{y} = \frac{\int yf(y)\,dy}{\int f(y)\,dy}$

Arc Length

$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$
$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}\,dt$
for parametric. Area is just this times $2\pi x$ or y for opposite of axis.

Vector Calculus

Cartesian

$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{x}} + dz\hat{\mathbf{z}}$, $d\tau = dxdydz$

Gradient

$\boldsymbol{\nabla}t = \partial_x t\hat{\mathbf{x}} + \partial_y t\hat{\mathbf{y}} + \partial_z t\hat{\mathbf{z}}$

Divergence

$\boldsymbol{\nabla}\cdot\mathbf{v} = \partial_x v_x\hat{\mathbf{x}} + \partial_y v_y\hat{\mathbf{y}} + \partial_z v_z\hat{\mathbf{z}}$

Curl

$\boldsymbol{\nabla}\times\mathbf{v} = (\partial_y v_z - \partial_z v_y)\,\hat{\mathbf{x}} +$
$(\partial_z v_x - \partial_x v_z)\,\hat{\mathbf{y}} + (\partial_x v_y - \partial_y v_x)\,\hat{\mathbf{z}}$

Laplacian

$\boldsymbol{\nabla}^2 t = \partial_{xx}t + \partial_{yy}t + \partial_{zz}t$

Spherical

$d\mathbf{l} = dr\hat{\mathbf{r}} + rd\theta\hat{\boldsymbol{\theta}} + r\sin\theta d\phi\hat{\boldsymbol{\phi}}$,
$d\tau = r^2\sin\theta dr d\theta d\phi$

Gradient

$\boldsymbol{\nabla}t = \partial_r t\hat{\mathbf{r}} + \frac{1}{r}\partial_\theta t\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\partial_\phi t\hat{\boldsymbol{\phi}}$

Divergence

$\boldsymbol{\nabla}\cdot\mathbf{v} =$
$\frac{1}{r^2}\partial_r(r^2 v_r) + \frac{1}{r\sin\theta}\partial_\theta(\sin\theta v_\theta) + \frac{1}{r\sin\theta}\partial_\phi v_\phi$

Curl

$\boldsymbol{\nabla}\times\mathbf{v} = \frac{1}{r\sin\theta}\left[\partial_\theta(\sin\theta v_\phi) - \partial_\phi v_\theta\right]\hat{\mathbf{r}} +$
$\frac{1}{r}\left[\frac{1}{\sin\theta}\partial_\phi v_r - \partial_r(rv_\phi)\right]\hat{\boldsymbol{\theta}} +$
$\frac{1}{r}\left[\partial_r(rv_\theta) - \partial_\theta v_r\right]\hat{\boldsymbol{\phi}}$

Laplacian

$$\nabla^2 t = \frac{1}{r^2}\partial_r(r^2\partial_r t) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta t) + \frac{1}{r^2\sin^2\theta}\partial_{\phi\phi} t$$

Cylindrical

$$d\mathbf{l} = ds\hat{\mathbf{s}} + sd\phi\hat{\boldsymbol{\phi}} + dz\hat{\mathbf{z}},\ d\tau = sdsd\phi dz$$

Gradient

$$\nabla t = \partial_s t\hat{\mathbf{s}} + \frac{1}{s}\partial_\phi t\hat{\boldsymbol{\phi}} + \partial_z t\hat{\mathbf{z}}$$

Divergence

$$\nabla\cdot\mathbf{v} = \frac{1}{s}\partial_s(sv_s) + \frac{1}{s}\partial_\phi(v_\phi) + \partial_z v_z$$

Curl

$$\nabla\times\mathbf{v} = \left[\frac{1}{s}\partial_\phi v_z - \partial_z v_\phi\right]\hat{\mathbf{s}} + \left[\partial_z v_s - \partial_s v_z\right]\hat{\boldsymbol{\phi}} + \frac{1}{s}\left[\partial_s(sv_\phi) - \partial_\phi v_s\right]\hat{\mathbf{z}}$$

Laplacian

$$\nabla^2 t = \frac{1}{s}\partial_s(s\partial_s t) + \frac{1}{s^2}\partial_{\phi\phi} t + \partial_{zz} t$$

Vector Identities

Triple Products

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})$$
$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B})$$

Product Rules

$$\nabla(fg) = f\nabla g + g\nabla f$$
$$\nabla(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\times(\nabla\times\mathbf{B}) + \mathbf{B}\times(\nabla\times\mathbf{A}) + (\mathbf{A}\cdot\nabla)\mathbf{B} + (\mathbf{B}\cdot\nabla)\mathbf{A}$$
$$\nabla\cdot(f\mathbf{A}) = f\nabla\cdot\mathbf{A} + \mathbf{A}\cdot(\nabla f)$$
$$\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B})$$
$$\nabla\times(f\mathbf{A}) = f\nabla\times\mathbf{A} + \mathbf{A}\times(\nabla f)$$
$$\nabla\times(\mathbf{A}\times\mathbf{B}) = (\mathbf{B}\cdot\nabla)\mathbf{A} - (\mathbf{A}\cdot\nabla)\mathbf{B} + \mathbf{A}(\nabla\cdot\mathbf{B}) + \mathbf{B}(\nabla\cdot\mathbf{A})$$

Second Derivatives

$$\nabla\cdot(\nabla\times A) = 0$$
$$\nabla\times(\nabla f) = 0$$
$$\nabla\times(\nabla\times\mathbf{A}) = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$$

Theorems

Gradient Theorem

$$\int_{\mathbf{a}}^{\mathbf{b}}(\nabla f)\cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a})$$

Divergence Theorem

$$\int_{\mathcal{V}}\nabla\cdot\mathbf{A}d\tau = \oint_{\mathcal{S}}\mathbf{A}\cdot d\mathbf{a}$$

Curl (Stokes') Theorem

$$\int_{\mathcal{S}}(\nabla\times\mathbf{A})\cdot d\mathbf{a} = \oint_{\mathcal{C}}\mathbf{A}\cdot d\mathbf{l}$$

Coordinate Transforms

Spherical

$$\begin{cases} x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ z &= r\cos\theta \end{cases} \qquad \begin{cases} \hat{\mathbf{x}} &= \sin\theta\cos\phi\hat{\mathbf{r}} + \cos\theta\cos\phi\hat{\boldsymbol{\theta}} - \sin\phi\hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} &= \sin\theta\sin\phi\hat{\mathbf{r}} + \cos\theta\sin\phi\hat{\boldsymbol{\theta}} + \cos\phi\hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} &= \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}} \end{cases}$$

$$\begin{cases} r &= \sqrt{x^2+y^2+z^2} \\ \theta &= \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right) \\ \phi &= \tan^{-1}\left(\frac{y}{x}\right) \end{cases} \qquad \begin{cases} \hat{\mathbf{r}} &= \sin\theta\cos\phi\hat{\mathbf{x}} + \sin\theta\sin\phi\hat{\mathbf{y}} + \cos\theta\hat{\mathbf{z}} \\ \hat{\boldsymbol{\theta}} &= \cos\theta\cos\phi\hat{\mathbf{x}} + \cos\theta\sin\phi\hat{\mathbf{y}} - \sin\theta\hat{\mathbf{z}} \\ \hat{\boldsymbol{\phi}} &= -\sin\phi\hat{\mathbf{x}} + \cos\phi\hat{\mathbf{y}} \end{cases}$$

Cylindrical

$$\begin{cases} x &= s\cos\phi \\ y &= s\sin\phi \\ z &= z \end{cases} \qquad \begin{cases} \hat{\mathbf{x}} &= \cos\phi\hat{\mathbf{s}} - \sin\phi\hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} &= \sin\phi\hat{\mathbf{s}} + \cos\phi\hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} &= \hat{\mathbf{z}} \end{cases}$$

$$\begin{cases} s &= \sqrt{x^2+y^2} \\ \phi &= \tan^{-1}\left(\frac{y}{x}\right) \\ z &= z \end{cases} \qquad \begin{cases} \hat{\mathbf{s}} &= \cos\phi\hat{\mathbf{x}} + \sin\phi\hat{\mathbf{y}} \\ \hat{\boldsymbol{\phi}} &= -\sin\phi\hat{\mathbf{x}} + \cos\phi\hat{\mathbf{y}} \\ \hat{\mathbf{z}} &= \hat{\mathbf{z}} \end{cases}$$